SAP PP MODULE

Production Order Management

A Practical Field Study on SAP Production Planning

Lifecycle of a Production Order: Creation · Release · Modification · Confirmation · Closure

Module	SAP Production Planning (SAP PP)
Subject Area	Production Order Management
Transaction Codes	CO01 | MD16 | CO02 | CO15
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SAP PP – Production Order Management | Academic Project Report

Executive Summary

This project report delivers an in-depth examination of the SAP Production Planning (SAP PP) module, specifically targeting the end-to-end lifecycle of Production Order management. As one of the most operationally critical modules within the SAP ERP ecosystem, SAP PP governs the entire manufacturing execution chain — from initial capacity planning and material requirement identification through to shop floor dispatch and order closure.

The study traces how a Production Order is initiated either manually via transaction CO01 or through the conversion of MRP-generated Planned Orders using transaction MD16. It then examines how orders are modified post-creation (CO02), confirmed upon physical production (CO15), and ultimately brought to Technical Completion (TECO) once manufacturing activities have concluded.

Each phase of the production order lifecycle is explained with detailed procedural steps, supported by annotated SAP GUI screenshots and practical navigation paths. The report also highlights critical integration touchpoints between SAP PP and other modules, including MM (inventory movements) and CO (cost settlement). A dedicated troubleshooting section addresses common errors encountered in real-world SAP PP implementations.

► Transaction CO01 — Enables direct creation of a Production Order with automatic BOM and Routing population.

► Transaction MD16 — Facilitates batch conversion of MRP-generated Planned Orders into

actionable Production Orders.

► Transaction CO02 — Supports modifications to quantity, scheduling, and master data in existing Production Orders.

► Transaction CO15 — Executes order confirmation, simultaneously triggering Goods Issue (Mv. Type 261) and Goods Receipt (Mv. Type 101).

► TECO Status — Signals manufacturing closure; freezes goods movements and removes the order from MRP consideration.

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Chapter 1: Introduction to SAP PP and the Production Order

SAP Production Planning (PP) is a foundational module within the SAP ERP system that manages every dimension of the manufacturing process. Its scope extends from long-range production scheduling and capacity utilization planning all the way down to real-time shop floor execution and material consumption tracking. Among all the functional objects within SAP PP, the Production Order stands as the central controlling document around which all manufacturing activities revolve.

1.1 What Is a Production Order?

A Production Order in SAP PP is a formal internal directive that authorizes and controls the manufacturing of a specific material. When a Production Order is created, it answers four critical manufacturing questions simultaneously:

- What to produce — the target material and its specifications.
- Where to produce it — the plant and production storage location.
- How much and when — the target quantity and scheduled dates.
- How to produce it — via the Bill of Materials (components) and Routing (operations at work centers).
- What it will cost — planned costs derived from component prices and routing activity rates.

An important behavioral distinction of Production Orders compared to Planned Orders is that Production Orders are static documents. Once created, they cannot be automatically revised by a subsequent MRP run. Any adjustments must be made manually through transaction CO02. This characteristic gives production planners stable, committed execution documents insulated from fluctuating MRP proposals.

1.2 From MRP Planned Order to Production Order

The Material Requirements Planning (MRP) engine in SAP automatically generates Planned Orders whenever it detects a material shortage for items with an internal procurement indicator. These Planned Orders represent procurement proposals — they are dynamic, adjustable, and reconsidered with every new MRP run. However, they cannot be used for shop floor execution.

To move from planning intent to manufacturing reality, Planned Orders must be converted into Production Orders. This conversion can be done individually for a single order or in bulk for a group of materials managed by the same MRP Controller, using transaction MD16. Upon conversion, the original Planned Order is permanently removed from the system and no longer participates in subsequent MRP calculations.

1.3 Master Data Automatically Copied Into a Production Order

When a Production Order is generated — whether through manual creation or Planned Order conversion — SAP automatically retrieves and populates it with the following master data:

► Bill of Materials (BOM): A complete, structured list of all raw materials, semi-finished goods, and sub-assemblies required to manufacture the finished product. The BOM drives component reservations and Goods Issue postings.

► Routing: A sequence of manufacturing operations defining what is done at each Work Center, along with machine time, labor time, and setup time parameters. The Routing drives scheduling and activity cost postings.

► Planned Costs: Automatically calculated based on the BOM component prices and routing activity prices. These provide a cost baseline against which actual costs are later measured during and after confirmation.

Chapter 2: Core Concepts and Key Terminology

Before engaging with the transactional steps involved in Production Order processing, it is essential to build a strong understanding of the foundational terminology and concepts that underpin the SAP PP module. These concepts directly govern system behavior and explain why each step in the process is necessary.

2.1 Foundational Concepts

Bill of Materials (BOM): A hierarchical, structured inventory of all components — raw materials, intermediate products, and packaging — required to manufacture a finished good. The BOM is automatically read and attached to the Production Order upon creation, forming the basis for component reservations and material consumption tracking.

Routing: A sequential description of the manufacturing operations that must be performed to complete the product. Each operation is assigned to a specific Work Center and carries time parameters (setup time, machine time, labor time) that drive production scheduling and activity cost calculations.

Backflushing: An automatic Goods Issue mechanism activated during order confirmation. Components with the Backflushing indicator set in their Material Master (MRP1 view) are automatically consumed from stock using Movement Type 261 without requiring manual goods issue postings.

Movement Type 101: The standard SAP movement type used for Goods Receipt (GR). When triggered during production confirmation, Movement Type 101 adds the produced quantity to the unrestricted-use stock of the finished or semi-finished material.

Movement Type 261: The standard movement type for Goods Issue (GI) from a production order. It reduces component inventory when raw materials and sub-assemblies are consumed during the manufacturing process.

TECO — Technical Completion: A lifecycle status applied to a Production Order when manufacturing is finished or the business decides not to continue executing the order. Once TECO status is set, no further goods movements can be posted against the order, it is removed from MRP, and all open component reservations are cancelled.

2.2 Transaction Code Reference

T-Code	Transaction Name	Functional Purpose
CO01	Create Production Order	Manually initiate a Production Order for a material at a defined plant, with BOM and Routing auto-populated.
MD16	Display & Convert Planned Orders	Review MRP-generated Planned Orders and convert selected entries into Production Orders in bulk.
CO02	Change Production Order	Modify an existing Production Order — update quantities, dates, component assignments, or set TECO status.
CO15	Confirm Production Order	Declare completed production quantities and trigger automatic Goods Receipt and Goods Issue postings.
CO03	Display Production Order	View a Production Order in read-only mode — useful for verification after conversion from a Planned Order.

Chapter 3: Creating and Releasing a Production Order (CO01)

Transaction CO01 is the starting point for initiating a Production Order directly within the SAP system. This approach is used when a planner or production controller wants to create a manufacturing order manually — outside of the automated MRP proposal mechanism. The order creation process simultaneously retrieves BOM and Routing master data, making it a comprehensive initialization step for shop floor execution.

An important configuration option worth noting is the ability to enable automatic order release during the creation transaction itself. This is configured by assigning a Production Scheduling Profile (with the Auto Release indicator activated) to the material in the Work Scheduling view of the Material Master. When this profile is active, the system releases the order immediately upon saving, bypassing the need for a separate release step.

3.1 Step-by-Step Navigation Guide

Step 1 — Launch CO01 and Enter Initial Parameters

Open transaction CO01 from the SAP Easy Access menu or by typing CO01 in the command field. The system displays the Production Order Create: Initial Screen. On this screen, enter the following mandatory fields:

1. Material Code — the SAP material number for the item to be manufactured.
2. Production Plant — the plant code where the order will be executed (e.g., SM01 – AVBV Enschede Mfg.).
3. Order Type — defines the controlling parameters for the order (typically PP01 for standard production).

Once the fields are populated, click the green checkmark or press Enter to proceed.

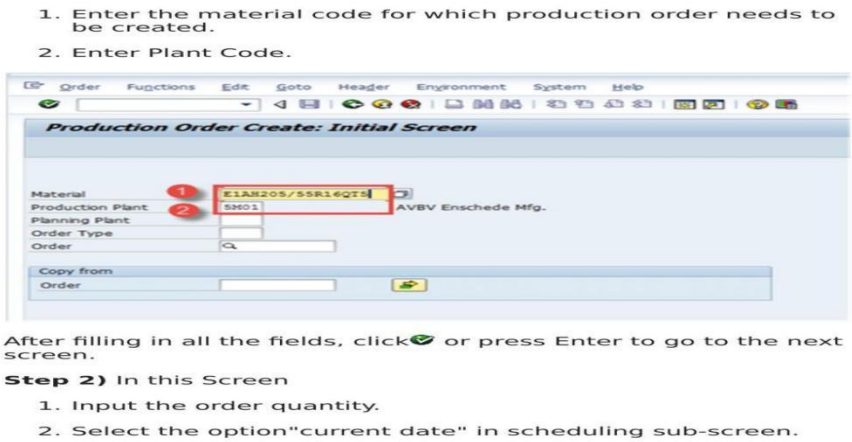


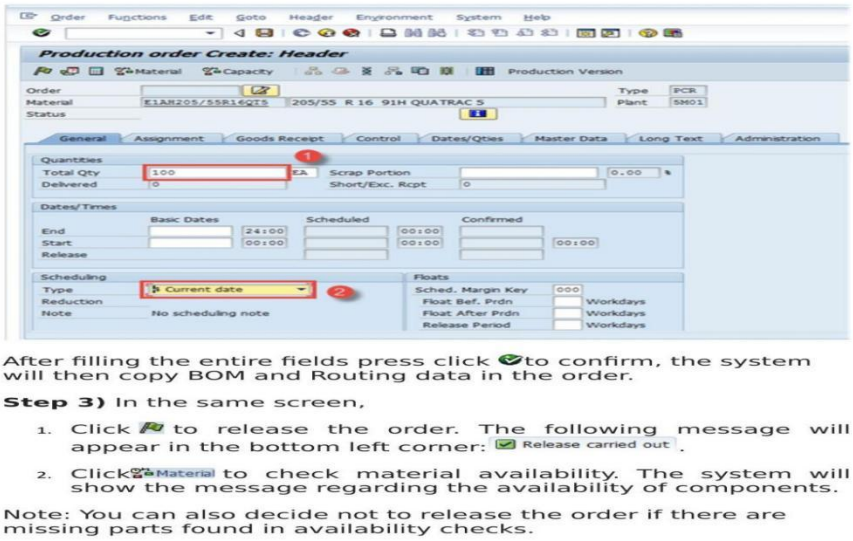
Figure 3.1 — CO01: Production Order Create: Initial Screen with Material and Plant Entry

Step 2 — Enter Quantity and Scheduling Parameters

On the Production Order Create: Header screen, complete the following details:

4. Total Quantity — specify the required production quantity and unit of measure (e.g., 100 EA).
5. Scheduling Type — select 'Current Date' in the Scheduling sub-section so the system calculates start and finish dates relative to today.

After filling in these fields, press the green checkmark to confirm. The SAP system will automatically read the valid BOM and Routing for the material and plant combination, copying their data into the order header and components tab.



After filling the entire fields press click to confirm, the system will then copy BOM and Routing data in the order.

Step 3) In the same screen,

1. Click to release the order. The following message will appear in the bottom left corner: Release carried out .
2. Click Material to check material availability. The system will show the message regarding the availability of components.

Note: You can also decide not to release the order if there are missing parts found in availability checks.

Figure 3.2 — CO01: Production Order Header displaying Quantity, Dates, and Scheduling Type

Step 3 — Release the Order and Verify Component Availability

With the order header populated, the next step is to release the order for shop floor execution:

6. Click the Release button (flag/release icon in the toolbar). The system will confirm with the status message 'Release carried out' in the bottom-left status bar.
7. Click the Material Availability Check button to verify that all required BOM components are available in sufficient quantity. The system will display a dialog indicating component availability or listing any missing parts.
8. Click Save. The system assigns a unique Production Order number and displays a confirmation message such as 'Order number XXXXXXXXXX saved' in the status bar.

Note: If the availability check reveals missing components, you may choose to save the order without releasing it. The order can be released later via CO02 once the missing parts are procured or transferred to the production storage location.

Chapter 4: Converting a Planned Order to a Production Order (MD16)

Transaction MD16 provides a powerful batch-conversion interface that allows production planners to review and selectively convert MRP-generated Planned Orders into Production Orders. This transaction is typically executed as part of the daily or weekly production planning routine, following an MRP run.

Planned Orders are inherently provisional — they are adjusted and regenerated each time MRP runs. Converting them to Production Orders commits the manufacturing intent, creates formal cost objects, and establishes component reservations on the shop floor. The original Planned Order is permanently deleted from the system upon conversion and will not appear in future MRP evaluations.

4.1 Step-by-Step Navigation Guide

Step 1 — Open MD16 and Select Search Method

Access transaction MD16 from the SAP Easy Access screen. The system presents the Display Planned Orders initial screen, offering several search criteria options. For bulk conversion by MRP area, select the MRP Controller radio button — this option retrieves all Planned Orders for a group of materials managed by a specific controller.

Confirm the selection by clicking the green checkmark or pressing Enter.

The system will show the screen as below.

Step 2) In this screen, we will fetch the planned orders based on MRP controller and selection dates.

1. Enter plant code.
2. Enter MRP controller
3. Enter end selection date till all planned orders will be extracted.

Click right mark sign from the bottom or enter from the keyboard to move to the next screen.

Step 3) In this screen,

Figure 4.1 — MD16: Display Planned Orders Initial Screen with MRP Controller Selection

Step 2 — Apply Filter Parameters

On the search parameter screen, specify the following filter criteria to retrieve the relevant Planned Orders:

9. Plant Code — identifies the manufacturing plant (e.g., PU – June Production).
10. MRP Controller — the controller code responsible for the group of materials (e.g., 010 SFG(EWH)).
11. End Selection Date — the system will retrieve all Planned Orders with a requirement date up to this cutoff (e.g., 31.10.2013).

Click the green checkmark to execute the search query.

Step 3 — Review and Convert Selected Orders

The system returns a list of all Planned Orders matching the filter criteria. The list displays key information including material number, planned quantity, order start and finish dates, and the assigned Planned Order number.

12. Review the list and identify the Planned Orders you wish to convert. Select the corresponding rows by clicking the checkboxes on the left side.
13. Click the 'Convert to Production Order' button in the toolbar. The system immediately creates Production Orders for all selected entries and permanently removes the source Planned Orders.

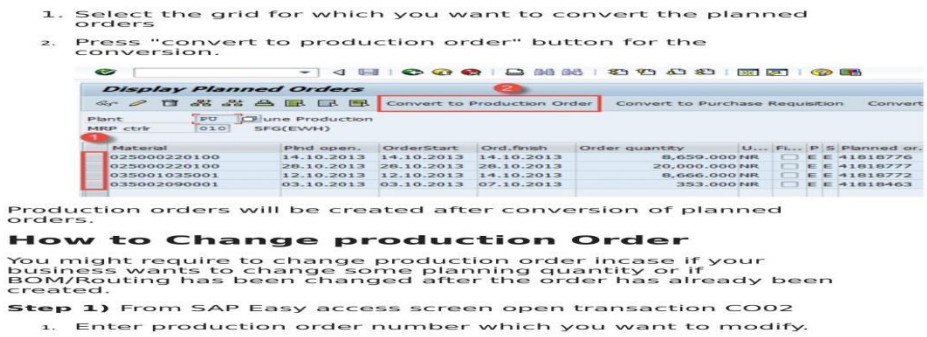


Figure 4.2 — MD16: Display Planned Orders List with 'Convert to Production Order' Button Highlighted

Note: Once a Planned Order has been converted, it is irreversibly deleted from the system. To view or verify the resulting Production Order, use transaction CO03 (Display Production Order) and search by the material or order number.

Chapter 5: Modifying an Existing Production Order (CO02)

After a Production Order has been created — whether manually or through Planned Order conversion — circumstances may arise that require changes to the order parameters. Common modification scenarios include adjustments to the production quantity due to demand revisions, date rescheduling, or the need to refresh BOM and Routing data after master data changes. Transaction CO02 is the standard SAP entry point for all such changes.

It is important to note that modifications made via CO02 are independent of MRP. The MRP engine cannot automatically update a Production Order once it has been created — planners must apply any necessary corrections manually.

5.1 Step-by-Step Navigation Guide

Step 1 — Open CO02 and Retrieve the Production Order

Launch transaction CO02 from the SAP Easy Access screen or via the command field. The system displays the Production Order Change: Initial Screen. Enter the relevant Production Order number in the Order field. Ensure the 'Display Overview' option is selected (it is pre-selected by default) and confirm by clicking the green checkmark or pressing Enter.

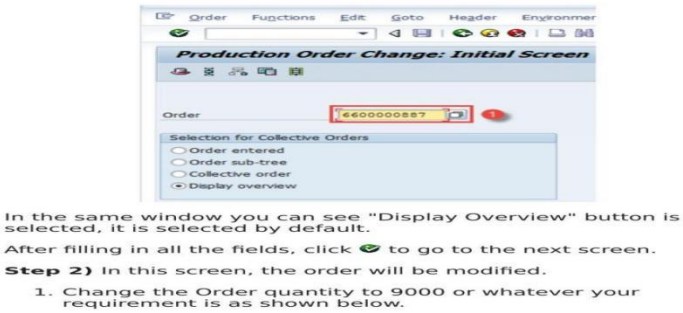


Figure 5.1 — CO02: Production Order Change Initial Screen with Order Number Entry

Step 2 — Apply Required Modifications

The system opens the Production Order Change: Header screen, where all editable fields are accessible. The most frequently updated field is the Total Quantity in the Quantities section — update this value to reflect

the revised production requirement (e.g., changing from 100 to 9,000 units). Other modifiable parameters include scheduling dates, component assignment details, order notes, and print control settings.

Once all changes have been applied, click the Save button. The system generates a confirmation message in the status bar, such as 'Order number 6600000887 saved,' indicating successful update.

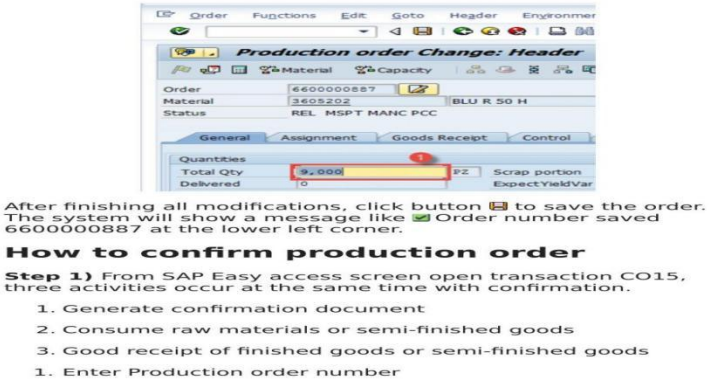


Figure 5.2 — CO02: Production Order Change Header Showing Updated Total Quantity to 9,000

Note: If the BOM or Routing has been updated in master data after the Production Order was created, the changes are not automatically reflected in the order. Use the Functions menu in CO02 to execute 'Read PP Master Data' — this refreshes the order's BOM and Routing from the current master data records.

Chapter 6: Confirming a Production Order (CO15)

Order confirmation is the critical transactional event that documents the completion of physical manufacturing activities. Executed through transaction CO15, the confirmation process simultaneously triggers three significant system-wide actions that update inventory, costs, and order status across multiple SAP modules.

#	Activity	System Impact
1	Confirmation Document	A formal record is generated capturing the yield quantity, scrap quantity, activity times, and the personnel involved in production.
2	Goods Issue (GI)	BOM components are automatically consumed from stock (Movement Type 261), reducing raw material or semi-finished goods inventory.
3	Goods Receipt (GR)	The finished or semi-finished goods are automatically posted to unrestricted-use stock (Movement Type 101), increasing available inventory.

6.1 Step-by-Step Navigation Guide

Step 1 — Open CO15 and Enter the Production Order Number

Launch transaction CO15 from the SAP Easy Access screen. The system displays the Enter Production Order Confirmation: Initial Screen. Enter the Production Order number in the Order field, then confirm by clicking the green checkmark or pressing Enter.

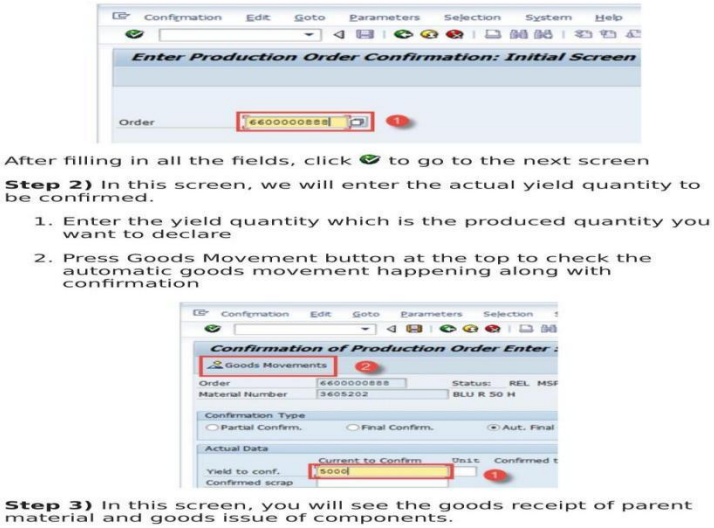


Figure 6.1 — CO15: Enter Production Order Confirmation Initial Screen with Order Number

Step 2 — Declare Yield and Review Goods Movements

The system advances to the Confirmation of Production Order screen. Complete the following fields:

14. Yield to Confirm — enter the actual quantity produced during this confirmation entry (e.g., 5,000 units).
15. Confirmation Type — select the appropriate type: Partial Confirmation (more production expected), Final Confirmation (all production declared complete), or Auto Final (system determines automatically).
16. Click the 'Goods Movements' button in the toolbar to preview all inventory postings that will be triggered upon saving this confirmation.

Step 3 — Validate and Save the Confirmation

The Goods Movements overview screen displays every automatic posting that will occur:

- Goods Receipt of the finished material with Movement Type 101 — stock of the produced item increases.
- Goods Issue of each component with Movement Type 261 — stock of consumed raw materials decreases.

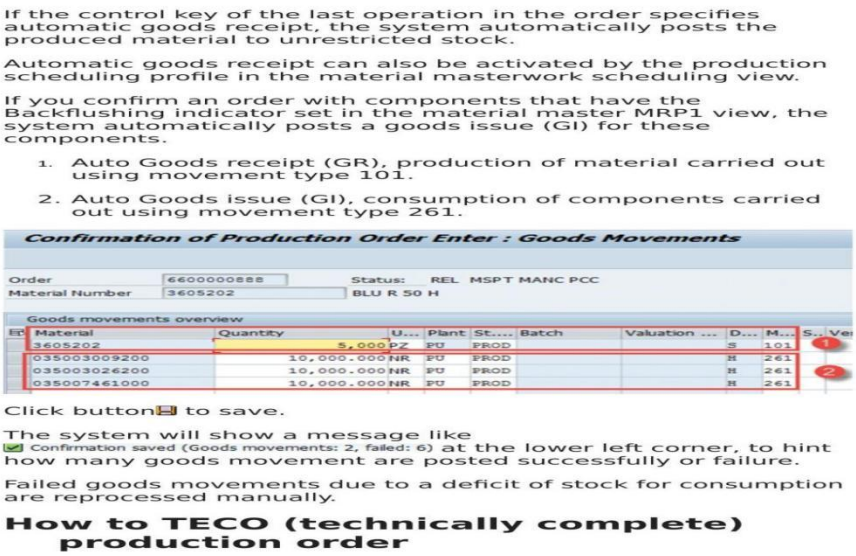


Figure 6.2 — CO15: Goods Movements Overview Showing GR (Mv.101) and GI (Mv.261) Postings

Click Save to post the confirmation. The system displays a summary in the status bar, such as 'Confirmation saved (Goods movements: 2, failed: 6),' indicating how many movements were processed successfully and how many failed due to stock deficits.

Note: Failed goods movements occur when a component does not have sufficient stock at the production storage location. These failed movements must be reprocessed manually after the missing components are either procured or transferred into the appropriate storage location.

6.2 Automatic Goods Movements During Confirmation

SAP PP can be configured to trigger goods movements automatically during the confirmation step, reducing manual intervention:

- Auto Goods Receipt (GR) is activated when the Control Key of the final operation in the Routing is configured to generate an automatic goods receipt. Alternatively, it can be enabled through the Production Scheduling Profile in the Material Master Work Scheduling view.
- Auto Goods Issue (GI) — Backflushing — is activated by setting the Backflushing indicator in the Material Master MRP1 view for each relevant component. When enabled, the system automatically posts the goods issue for these components during confirmation using Movement Type 261.

6.3 Cost Flow During the Confirmation Process

The confirmation step is also a critical financial event in SAP Controlling (CO). Understanding the cost flow helps avoid confusion during period-end cost settlement:

- Activity Costs — Machine time costs, labor costs, and setup costs are debited to the Production Order based on the confirmed activity quantities and the activity prices defined in Cost Center Accounting.
- Component Costs — The cost of raw materials and semi-finished goods is debited to the Production Order at the time of Goods Issue (Movement Type 261).
- Goods Receipt Credit — When the produced material is received into stock (Movement Type 101), the Production Order is credited based on the standard cost of the finished material. Any remaining debit balance represents production variance.

Chapter 7: Applying Technical Completion (TECO) Status

Technical Completion (TECO) is the final lifecycle milestone for a Production Order in SAP PP. Applying TECO status formally signals that shop floor activities for the order have either been fully executed or will not be carried out further. This status change has far-reaching consequences across multiple SAP modules and is a prerequisite for cost variance analysis and period-end closing activities.

7.1 When to Apply TECO Status

TECO should be applied in the following scenarios:

- The Production Order has been fully confirmed and all produced quantities have been received into inventory — a standard post-production closure step.
- The business has made a decision to cancel or halt production on the order, even if only partial quantities have been produced.
- Period-end closing activities require all completed orders to be brought to TECO status so that the Controlling team can perform variance calculations.

7.2 System-Wide Effects of TECO Status

Once TECO is applied to a Production Order, the following system-wide changes are enforced immediately:

► Order Status: The system status field updates to TECO — no further Goods Issues or Goods Receipts can be posted against the order.

► MRP Exclusion: The order is removed from the stock/requirements list and will no longer influence future MRP planning runs.

► Reservation Cancellation: All open component reservations tied to the Production Order are automatically deleted, freeing up that stock for other requirements.

► Controlling Enablement: The Controlling (CO) department can now execute variance calculations to compare planned vs. actual costs on the order.

7.3 Step-by-Step Navigation Guide

Step 1 — Open CO02 and Retrieve the Target Order

TECO is applied through transaction CO02 (Change Production Order). Open CO02, enter the Production Order number to be technically completed, and click the green checkmark to navigate to the order header screen.

Technical completion of the order is required when production order is either fully delivered, or it is not to be processed further at shop floor.

After technical completion order status changes to TECO, no further goods movement can be done against the order.

Order is no longer available in MRP run and gets deleted from stock/requirement list.

The controlling department can then carry out the variance calculation on the production Order after TECO status.

Step 1) From SAP easy access, open transaction CO02

1. Enter the order number which you want to TECO.

Click to next screen after filling all fields,

Step 2) In this screen, we will see how to place production order to TECO status,

1. Click in menu Function-> Restrict processing ->complete technically System will show message in status- TECO

Figure 7.1 — CO02: Production Order Change Initial Screen — Entry Point for TECO Application

Step 2 — Navigate to Complete Technically via Functions Menu

On the Production Order Change: Header screen, access the menu path: Functions → Restrict Processing → Complete Technically. The system immediately updates the order status. The Status field in the order header will now display: TECO MSPT MANC NTUP PCC, confirming that technical completion has been applied.

Save the order to permanently commit the TECO status.

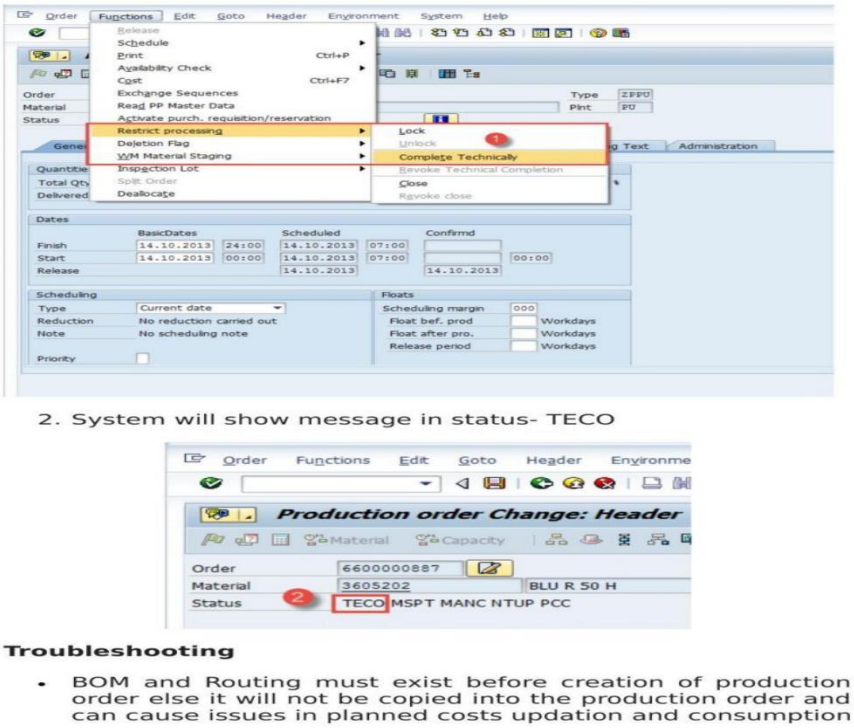


Figure 7.2 — CO02: Functions Menu showing 'Complete Technically' Option and Resulting TECO Status in Order Header

Note: TECO can be reversed using the 'Revoke Technical Completion' option in the same Functions → Restrict Processing menu, if goods movements or further confirmations are needed after TECO was set in error.

Chapter 8: Common Issues and Troubleshooting Guide

Production Order processing in SAP PP involves multiple integration points with master data, inventory management, and costing. Issues encountered during day-to-day operations are often traceable to configuration gaps or missing master data. The following table documents the most frequently encountered problems and their recommended remedies.

Symptom / Error	Root Cause	Recommended Resolution
BOM or Routing data not populated in Production Order	BOM or Routing does not exist or is not valid for the material/plant combination.	Verify that a valid BOM exists in transaction CS03 and a valid Routing exists in CA03 for the material and plant. Create or extend the master data as needed, then recreate the Production Order.
Production Order cannot be confirmed (CO15)	The order status is not 'Released.' Confirmation requires REL status.	Open the order in CO02, navigate to Functions → Release to manually release it, or enable auto-release via the Production Scheduling Profile in the Material Master.
Accounting error during Goods Movement confirmation	Incorrect or missing Valuation Class in the Material Master Accounting view.	Check the Valuation Class in the Material Master (MM03 – Accounting 1 view). Coordinate with the Finance/CO team to assign or correct the Valuation Class.
Activity price or costing error on confirmation	Standard cost not updated or cost estimate not released for the material.	Execute a cost estimate for the material in CK11N and release it with CK24. This ensures planned costs are available for activity price calculation.
Goods movement failed due to stock deficit	Insufficient component stock at the production storage location.	Procure or transfer the missing components to the correct storage location using a stock transfer order or inbound delivery. Then reprocess the failed goods movements.
Planned Order not visible after conversion	The Planned Order is permanently deleted from the system after conversion.	Use CO03 (Display Production Order) to locate the resulting Production Order by material, plant, or order number.

Chapter 9: Conclusion and Key Learnings

This report has provided a structured, end-to-end walkthrough of Production Order management within the SAP Production Planning module. Beginning with the conceptual underpinnings of what a Production Order represents and how it integrates with master data, the study then examined each transactional milestone in the order lifecycle with practical, step-by-step guidance supported by system screenshots.

9.1 Summary of Key Takeaways

- Production Orders serve as the backbone of SAP PP execution — they carry BOM, Routing, scheduling, and cost data, driving material consumption, inventory movements, and financial postings.
- The lifecycle begins either through manual creation via CO01 or through the structured conversion of MRP-proposed Planned Orders via MD16 — each pathway suited to different planning scenarios.
- Transaction CO02 provides the flexibility to modify orders post-creation and also serves as the gateway to TECO status, giving production planners full lifecycle control over each manufacturing document.
- Order confirmation via CO15 is the most operationally significant step — it records actual production, triggers automatic inventory movements, and updates the order with activity and component costs.
- TECO status provides a formal and auditable mechanism to close the production cycle, enabling Controlling teams to measure cost variance and finalize period-end reporting.
- Accurate master data — BOM, Routing, Standard Cost Estimate, and Valuation Class — is the single most important prerequisite for error-free Production Order processing across all transaction codes.

9.2 Business Value of SAP PP Mastery

Mastering the Production Order lifecycle in SAP PP translates directly to tangible operational improvements. Organizations that execute production orders efficiently — with correct master data, automated goods movements, and timely TECO closure — benefit from real-time inventory accuracy, tighter cost control, reduced manual intervention, and faster financial period-end closing cycles.

For students and aspiring SAP PP consultants, a thorough understanding of CO01, MD16, CO02, CO15, and the TECO process forms the practical competency core around which all advanced PP topics — capacity planning, production versions, schedule agreements, and repetitive manufacturing — are layered.

► CO01: Always verify BOM and Routing availability before creating a Production Order to avoid missing cost and component data.

► MD16: Establish a consistent batch conversion schedule aligned with MRP run frequency to ensure timely Production Order creation.

► CO02: Use 'Read PP Master Data' after any BOM or Routing change to keep Production Orders

synchronized with current master data.

► CO15: Review the Goods Movements overview before saving to catch potential stock deficits early and avoid failed postings.

► TECO: Apply Technical Completion promptly after final delivery to prevent open orders from distorting MRP planning calculations.

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